

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Advanced NLP, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Publication

- “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023** (3rd International Conference on Intelligent Technologies), applied deep transfer learning on neuroimaging data for medical image classification (first author).

Skills

Languages	Python, C++, JavaScript, TypeScript, SQL, Bash
Deep Learning & Generative	PyTorch, JAX, HuggingFace Transformers, NumPy, diffusion (DDPM), normalizing flows, CNNs / ResNet, transfer learning, variational inference, from-scratch autodiff
Vision & Multimodal	image classification / computer vision, medical-image transfer learning, multimodal pipelines (vision + speech + text), MediaPipe, whisper.cpp
Retrieval / QA / Research	RAG, multi-hop QA, RRF, cross-encoder reranking, MMR, sentence-transformers, mechanistic interpretability, evaluation methodology, statistical analysis (95% CIs)
Cloud & Tools	AWS, GCP, Azure, Docker, Git, GitHub Actions, Linux, Ollama, spaCy

Research & Projects

Generative Models & Deep Learning

github.com/Deva-1903/UMASS_CICS_689

JAX, PyTorch, NumPy | UMass CS 689 Advanced Machine Learning

- Implemented **generative models from scratch** in JAX, a **DDPM diffusion model** and a **normalizing flow** with coupling layers, deriving the ELBO and closed-form Gaussian-KL objectives via variational inference.
- Trained and benchmarked **5 neural architectures** (perceptron, MLPs, VGG-style CNN, ResNet) across 3 optimizers on **CIFAR-10** image classification, and built a matrix-based **reverse-mode autodiff engine from scratch in NumPy**, validated against JAX to machine precision.

Sonare - Multimodal Vision + Speech Desktop Application

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

MediaPipe, whisper.cpp, FastAPI, React/Electron | Qualcomm Edge AI Hackathon 2025

- Built an on-device **multimodal** pipeline fusing **vision** (MediaPipe hand tracking, gesture classification), **speech** (whisper.cpp ASR), and **text** (local Text-To-Speech), integrating camera and microphone inputs end-to-end on device.
- Engineered async APIs and queues to keep the UI responsive while processing live video and audio streams locally, fully offline with no external network dependencies.

KG2RAG-Enhanced - Multi-Hop Retrieval, QA & Reasoning

github.com/Deva-1903/KG2RAG-685-NLP

Python, Ollama + LLaMA-3, sentence-transformers, llama-index, spaCy | UMass CS 685 NLP

- Led the **multi-view retrieval** of a team knowledge-grounded **RAG** pipeline (sub-question decomposition, per-view embedding retrieval, RRF fusion, cross-encoder reranking, MMR), lifting supporting-fact recall **+2.68%** (54.53 to 57.21) over the KG²RAG baseline on a **4,905-question** evaluation with 95% confidence intervals.

Experience

Wysa

July 2023 – July 2025

Backend Engineer

Bengaluru, India

- Architected full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label data for production models; replaced spreadsheet workflows and accelerated iteration on downstream NLP/clinical models.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability and remediated **20+ VAPT findings**.

Extracurriculars

- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.